MODEL INSTITUTE OF ENGINEERING & TECHNOLOGY, JAMMU

# SilkSphere

Project Objectives & Team Responsibilities

Minor Project Briefing — Department of Computer Science and Engineering

## Objectives

**1. Disease Detection —** Develop and deploy an AI-based system to detect silkworm disease (Grasserie) by classifying worms as Healthy vs Infected, enabling early isolation before it spreads.

**2. IoT Integration —** Set up sensors to monitor rearing conditions — temperature, humidity, and CO2 — in real time and automatically correct unsafe conditions.

**3. Live Dashboard —** Provide a real-time dashboard for farmers and buyers showing live sensor data, disease-detection results, and batch status.

**4. Marketplace —** Build a marketplace where farmers can list and sell their cocoon batches directly to buyers, removing middlemen.

## Team Responsibilities

| Member | Role & Contribution |
|---|---|
| **Himanshu Chib** | UI/UX design; research on silkworm environmental conditions; support in IoT integration; support in AI model training |
| **Ujval Singh** | IoT setup, design, and integration; support in AI model training |
| **Simriti Pandita** | Disease detection model training; marketplace integration; support in IoT integration |

Submitted by:
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Reviewed by (Mentor):